

W5D2 Output Screenshots

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C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\Sanjana CA>ollama --version
ollama version is 0.32.5

C:\Users\Sanjana CA>ollama pull llama3.2:3b
pulling manifest
pulling dde5aa3fc5ff: 100% 2.0 GB
pulling 966de95ca8a6: 100% 1.4 KB
pulling fcc5a6bec9da: 100% 7.7 KB
pulling a70ff7e570d9: 100% 6.0 KB
pulling 56bb8bd477a5: 100% 96 B
pulling 34bb5ab01051: 100% 561 B
verifying sha256 digest
writing manifest
success

C:\Users\Sanjana CA>ollama run llama3.2:3b
>>> What is Artificial Intelligence?
Artificial Intelligence (AI) refers to the development of
computer systems that can perform tasks that typically
require human intelligence, such as:

1. Learning: AI systems can learn from data and improve
their performance over time.
2. Problem-solving: AI systems can solve complex problems
and make decisions based on available data.
3. Perception: AI systems can interpret and understand
data from sensors, such as images and speech.
4. Reasoning: AI systems can draw conclusions and make
inferences based on available data.

The goal of AI is to create computer systems that can
think and act like humans, with the ability to:

1. Understand and interpret data
```

```
C:\WINDOWS\system32\cmd. x + v
1. Understand and interpret data
2. Make decisions and take actions
3. Learn from experience
4. Adapt to new situations

There are several types of AI, including:

1. Narrow or Weak AI: Designed to perform a specific task,
such as image recognition or language translation.
2. General or Strong AI: A hypothetical AI system that
possesses human-like intelligence and can perform any
intellectual task.
3. Superintelligence: An AI system that surpasses human
intelligence in all domains.

AI has many applications, including:

1. Virtual assistants, such as Siri and Alexa
2. Image and speech recognition systems
3. Self-driving cars and drones
4. Predictive maintenance and quality control
5. Personalized recommendations and customer service

The benefits of AI include:

1. Increased efficiency and productivity
2. Improved decision-making and problem-solving
3. Enhanced user experience and convenience
4. Economic growth and competitiveness

However, AI also raises concerns about:

1. Job displacement and automation
2. Bias and discrimination in AI systems
3. Security and privacy risks
4. Dependence on data quality and availability
```

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C:\WINDOWS\system32\cmd. x + v

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2. Bias and discrimination in AI systems
3. Security and privacy risks
4. Dependence on data quality and availability

Overall, AI is a rapidly evolving field that has the
potential to transform industries and improve lives, but
it also requires careful consideration of its potential
risks and consequences.

>>> Send a message (/? for help)
```

```
Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama> dir

Directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama

Mode                LastWriteTime         Length Name
----                -
d-----          04-08-2026      19:46             screenshots
-a---l          04-08-2026      19:18             0 model_comparison.py
-a---l          04-08-2026      19:34          718 ollama_system_prompt.py
-a---l          04-08-2026      19:18             0 prompts.txt
-a---l          04-08-2026      19:19             0 README.md

PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama> python ollama_system_prompt.py
=====
Prompt 1
What is Machine Learning?
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Machine Learning is a way to teach computers to learn from data. It helps computers make good choices without being told exactly what
to do.

Imagine you have a big box of toys, and you want a robot to sort them into boxes by color. The robot looks at many examples of each t
oy, like "this red block goes in the red box". Then it learns to put similar blocks together. Over time, the robot gets better and be
tter at sorting the toys!

Machine Learning does something similar with big amounts of data.
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Prompt 2
Explain Neural Networks.
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Neural networks are like computers that think.
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Windows PowerShell
Prompt 2
Explain Neural Networks.
-----
Neural networks are like computers that think.

They have many layers, like a stack of blocks. Each layer looks at the information and says "yes" or "no".

The "blocks" learn from each other. When one says "yes", another says "more". This makes them better at solving problems.

Imagine a robot learning to recognize pictures. It sees many pictures and says "cat" when it sees a cat picture. The more it sees, the better it gets!

Neural networks are used in many things like computers, smartphones, and even cars. They help make them smarter.
=====
Prompt 3
What is Prompt Engineering?
-----
Prompt engineering is making special messages to help computers understand us better. It's like writing a clear letter to the computer.

We want the computer to get what we mean, so we make sure the message is short, simple, and easy to read. We try to use words that the computer can understand.

Think of it like giving a map to someone who lost their way. The map helps them find where they need to go. In this case, the map is our special message (prompt) that helps the computer get what we want from it.
=====
Prompt 4
What is Deep Learning?
-----
Deep learning is a way to make computers learn from pictures, sounds, and other things like humans. It's called "deep" because it uses many layers of computer code to look at things. These layers help the computer understand what it sees.

Imagine you're trying to teach a child to recognize cats. You show them a picture of a cat and say "this is a cat". Next time, you show them another picture and say "this is also a cat". The computer does the same thing with many pictures, so it can learn to recognize what's in each one.

Deep learning helps computers see things like humans do!
```

```
Windows PowerShell
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Deep learning helps computers see things like humans do!
=====
Prompt 5
Difference between AI and ML?
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AI (Artificial Intelligence) is like a computer that can think. It does things for us, like play games or understand voices.

ML (Machine Learning) is like a special kind of brain for computers. It helps the computer learn from data, so it can do new things without being told.

Think of AI like a robot, and ML like a tool inside the robot that makes it smarter.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_ollama>
```

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Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama> python model_comparison.py
=====
QUESTION: Explain Artificial Intelligence.

MODEL: llama3.2:3b
Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and
mimic their actions. The term may also be applied to any machine that exhibits traits associated with a human mind such as learning a
nd problem-solving.

At its core, AI involves two main components:

1. Machine Learning: This is a subset of AI that involves the use of algorithms and statistical models to enable machines to learn
from data, identify patterns, and make decisions without being explicitly programmed.
2. Natural Language Processing (NLP): This is another subset of AI that deals with the interaction between computers and humans i
n natural language, enabling computers to understand, interpret, and generate human language.

AI has numerous applications across various industries, including:

1. Virtual Assistants: AI-powered virtual assistants like Siri, Alexa, and Google Assistant can perform tasks such as answering q
uestions, setting reminders, and controlling smart home devices.
2. Image Recognition: AI algorithms can be used to identify objects, people, and patterns in images, enabling applications such a
s facial recognition, object detection, and self-driving cars.
3. Speech Recognition: AI-powered speech recognition systems can transcribe spoken words into text, enabling applications such as
voice-to-text functionality and language translation.
4. Expert Systems: AI-powered expert systems can mimic the decision-making abilities of human experts in specific domains, enabli
ng applications such as medical diagnosis and financial analysis.

The benefits of AI include:

1. Increased Efficiency: AI can automate repetitive tasks, freeing up human resources for more strategic and creative work.
2. Improved Accuracy: AI can reduce errors and improve accuracy in tasks such as data analysis and image recognition.
3. Enhanced Customer Experience: AI-powered chatbots and virtual assistants can provide 24/7 customer support and personalized re
commendations.
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```



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Windows PowerShell
ns include chatbots, virtual assistants, and document translation services.

5. Ethical Considerations:
  - AI has the potential to impact privacy, employment, decision-making processes, and more. Therefore, ethical considerations are crucial in its development and application.

6. Challenges:
  - Developing reliable and robust systems that can handle complex problems without errors or biases.
  - Ensuring fairness and accountability in AI-driven decisions.

AI is rapidly evolving, with significant progress being made in these areas, influencing how we live, work, learn, and interact with technology.
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QUESTION: What is Prompt Engineering?

MODEL: llama3.2:3b
Prompt Engineering is a subfield of Natural Language Processing (NLP) that focuses on designing, optimizing, and refining the inputs (prompts) used to elicit specific responses from language models or other AI systems. The goal of prompt engineering is to improve the performance, accuracy, and relevance of the generated text.

In traditional NLP tasks, prompts are often treated as a simple input parameter, without much consideration for their impact on the model's behavior. However, in recent years, researchers have recognized the importance of carefully designing and tuning prompts to achieve better results.

Prompt engineering involves several key aspects:

1. Understanding the task: Identifying the specific task or objective that needs to be achieved through prompt engineering.
2. Designing effective prompts: Crafting well-formed, informative, and concise prompts that elicit desired responses from the model.
3. Testing and iteration: Testing the prompts with the model, analyzing the results, and refining the prompts based on feedback.
4. Optimization techniques: Applying various optimization techniques, such as beam search or reinforcement learning, to improve prompt performance.

Effective prompt engineering can lead to several benefits, including:

1. Improved accuracy: Better responses from language models, reducing errors and inconsistencies.
2. Increased efficiency: More efficient use of model resources, leading to faster processing times and improved throughput.
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Windows PowerShell
3. Enhanced user experience: More engaging, relevant, and informative text generated by the model.

Some common techniques used in prompt engineering include:

1. Prompt crafting: Using linguistic and cognitive principles to design effective prompts that take into account the context, intent, and nuances of human language.
2. Attention mechanisms: Focusing attention on specific parts of the input or output to improve performance.
3. Meta-learning: Developing models that can adapt to new tasks and domains with minimal retraining.

By applying these techniques and principles, prompt engineers aim to create more effective, efficient, and user-friendly language models that generate high-quality responses to a wide range of prompts.

MODEL: qwen2.5:3b
Prompt Engineering refers to the process of fine-tuning and optimizing prompts (the input text given to large language models) to achieve better outputs or results from them. It involves understanding how different prompt variations can influence the generated responses, and it aims at making the interaction more effective, efficient, or aligned with specific goals.

Key aspects of Prompt Engineering include:

1. Understanding Model Capabilities: Knowing what your model is good at (its strengths) and where it might falter or be less accurate.
2. Tailoring Prompts to Specific Goals: Crafting prompts that align with the task or question you are trying to solve, whether for clarification, persuasion, storytelling, etc.
3. Ensuring Diversity of Responses: Using a variety of techniques such as providing different types of questions, using synonyms, varying sentence structures, or changing the context slightly can help in generating diverse responses from models that might be more versatile and helpful.
4. Context Management: Ensuring the model has all necessary context to understand the input correctly. This can involve adding relevant background information or clarifying instructions.
5. Error Handling: Being prepared for potential errors, such as overly long prompts which could cause a prompt timeout or too many iterations in case of complex models like Qwen.

By mastering Prompt Engineering, one can improve communication with AI systems and leverage their full potential effectively. It's not just about crafting better questions to ask the model; it's also about understanding how different ways of framing those questions can result in more tailored, accurate, or creative outputs from the system.
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QUESTION: Why is Python popular?
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Windows PowerShell
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QUESTION: Why is Python popular?

MODEL: llama3.2:3b
Python is a popular programming language due to several reasons:

1. **Easy to Learn**: Python has a simple syntax and is relatively easy to read and write, making it an ideal language for beginners.
2. **Versatile**: Python can be used for a wide range of applications, including web development, data analysis, machine learning, automation, and more.
3. **Large Community**: Python has a large and active community, with many libraries, frameworks, and tools available for various tasks.
4. **Cross-Platform**: Python can run on multiple operating systems, including Windows, macOS, and Linux.
5. **Fast Development**: Python's syntax and nature make it ideal for rapid prototyping and development.
6. **Extensive Libraries**: Python has a vast collection of libraries and frameworks that provide pre-built functionality, making it easy to get started with specific tasks.
7. **Dynamic Typing**: Python is dynamically typed, which means you don't need to declare the data type of a variable before using it.
8. **Object-Oriented**: Python supports object-oriented programming (OOP) concepts like classes, objects, inheritance, and polymorphism.
9. **Open-Source**: Python is open-source software, which means it's free to use and distribute.
10. **Career Opportunities**: Python is a highly sought-after skill in the job market, making it an excellent choice for those looking to start or advance their careers.

Some of the most popular applications of Python include:

* Data analysis and science (e.g., NumPy, pandas, scikit-learn)
* Web development (e.g., Django, Flask)
* Machine learning and artificial intelligence (e.g., TensorFlow, Keras)
* Automation and scripting (e.g., PyAutoGUI, paramiko)
* Education and research

Overall, Python's popularity can be attributed to its versatility, ease of use, and the large community of developers who contribute to its ecosystem.

MODEL: qwen2.5:3b
Python has gained immense popularity for several key reasons:
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Windows PowerShell

1. **Ease of Learning**: Python was designed with simplicity in mind, making it one of the easiest programming languages to learn and understand. Its readability and concise syntax make programs much easier to read compared to other more complex languages.

2. **Versatility**: It can be used for a wide range of applications including web development, data analysis, artificial intelligence, machine learning, scientific computing, software automation, and many others. This versatility makes Python suitable for almost any programming need.

3. **Large Standard Library and Third-party Modules**: Python comes with an extensive standard library that covers common use cases out-of-the-box. Additionally, there are countless third-party modules available on platforms like PyPI (Python Package Index), which allows developers to find and use a broad range of tools and libraries for specific tasks.

4. **Active Community Support**: Python has one of the most active open-source communities globally. This means that users can easily find support online through forums, documentation, and community-driven projects. Moreover, many educational institutions incorporate Python into their curriculum due to its simplicity and versatility, which contributes to a large base of developers who are proficient in it.

5. **Cross-Platform Compatibility**: Python's code is platform-independent (cross-platform). This means that the same code can run on different platforms like Windows, Linux, macOS without needing any modification, making development easier and more efficient.

6. **Career Opportunities**: The demand for skilled Python developers has grown significantly due to its widespread use in various industries. It offers excellent career opportunities both as a standalone programming language and as a tool within many other software environments.

7. **Robust Ecosystem of Tools**: Python boasts an array of powerful tools, such as Jupyter Notebooks (for data science), TensorFlow or PyTorch for machine learning, Flask and Django for web development, and more, which enhance productivity and ease of use.

These factors combined make Python one of the most popular programming languages today.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama> python model_comparison.py
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QUESTION: Explain Artificial Intelligence.

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Windows PowerShell
1. Learning: AI systems can learn from data and improve their performance over time.
2. Problem-solving: AI systems can solve complex problems by analyzing data and making decisions based on that analysis.
3. Perception: AI systems can interpret and understand data from sensors and other sources, such as images and speech.
4. Reasoning: AI systems can draw conclusions and make decisions based on the information they have been trained on.

There are several types of AI, including:

1. Narrow or Weak AI: This type of AI is designed to perform a specific task, such as facial recognition or language translation.
2. General or Strong AI: This type of AI is designed to perform any intellectual task that a human can, and has the ability to reason, learn, and apply knowledge across a wide range of tasks.
3. Superintelligence: This type of AI is significantly more intelligent than the best human minds, and has the potential to solve complex problems that are currently unsolvable.

AI systems typically use algorithms and data structures to process information, and can be trained using a variety of techniques, including:

1. Machine learning: This involves training an AI system on large amounts of data, so it can learn to recognize patterns and make predictions.
2. Deep learning: This is a type of machine learning that uses neural networks to analyze complex data sets.
3. Rule-based systems: These are AI systems that use pre-defined rules to make decisions.

AI has many applications in various fields, including:

1. Virtual assistants: AI-powered virtual assistants, such as Siri and Alexa, can perform tasks such as setting reminders and sending messages.
2. Image recognition: AI-powered image recognition systems can identify objects and people in images.
3. Natural language processing: AI-powered natural language processing systems can analyze and understand human language.
4. Robotics: AI-powered robots can be programmed to perform a variety of tasks, such as assembly and maintenance.

However, AI also raises several challenges and concerns, including:

1. Job displacement: The increasing use of AI in the workplace could displace jobs that are performed by humans.
2. Bias: AI systems can perpetuate biases if they are trained on biased data sets.
3. Security: AI-powered systems can be vulnerable to cyber attacks.
4. Ethics: AI raises several ethical concerns, such as the potential for AI systems to make decisions that affect human lives.

Overall, Artificial Intelligence has the potential to transform many aspects of our lives, but it also requires careful consideration
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Windows PowerShell

Overall, Artificial Intelligence has the potential to transform many aspects of our lives, but it also requires careful consideration and regulation to ensure that its benefits are realized while minimizing its risks.

MODEL: qwen2.5:3b
Artificial Intelligence (AI) is a field of computer science that aims to create systems and software capable of performing tasks that typically require human intelligence. These tasks include learning from experience, understanding language, using context, solving problems, drawing conclusions, making decisions based on new inputs, or recognizing patterns. AI can be categorized into different types based on its capabilities:

1. **Weak (Narrow) AI**: This type of AI is designed to perform specific tasks and functions as they have been explicitly programmed for certain situations. Examples include speech recognition software that understands only the specific language it has been trained with.

2. **Strong (General) AI**: Also known as Artificial General Intelligence (AGI), this type of AI would be capable of understanding, learning, and solving any intellectual problems that a human being can solve. At present, AGI is still theoretical and does not exist as a functional technology.

AI systems are often used in applications such as:

- **Machine Learning**: A subset of AI where machines can learn from data without explicit programming.
- **Deep Learning**: A type of machine learning inspired by the structure and function of the human brain's neural networks, which is highly effective for tasks like image and speech recognition.
- **Natural Language Processing (NLP)**: Allows computers to understand, interpret, and generate human language. This technology can be used in chatbots, virtual assistants, or automated customer support systems.

The development of AI has numerous applications across various industries such as healthcare, finance, manufacturing, transportation, entertainment, and many more. However, it also raises ethical concerns about job displacement, privacy issues, bias in decision-making processes, and the potential for misuse (e.g., autonomous weapons). As AI continues to evolve, so do our responsibilities and challenges surrounding its use.
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QUESTION: What is Prompt Engineering?

MODEL: llama3.2:3b
Prompt Engineering is the process of designing, testing, and refining input prompts that elicit specific responses or actions from AI models, such as language models, chatbots, and other conversational interfaces. The goal of prompt engineering is to optimize the performance of these systems by crafting high-quality inputs that can accurately capture user intent, reduce errors, and improve overall system usability.
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Windows PowerShell

Prompt Engineering involves a combination of natural language processing (NLP) expertise, linguistics, and machine learning knowledge . It requires understanding how AI models process and generate text, as well as the nuances of human communication. By carefully designing prompts, prompt engineers can help to:

1. Clarify user intent: Ensure that the AI model understands what the user is trying to accomplish or ask.
2. Reduce errors: Minimize the number of incorrect responses or actions generated by the AI model.
3. Improve accuracy: Increase the accuracy of the AI model's responses and improve overall performance.
4. Enhance user experience: Create prompts that are engaging, concise, and easy to understand.

Some key aspects of Prompt Engineering include:

1. Understanding the model's capabilities and limitations
2. Analyzing human communication patterns and language structures
3. Developing clear and concise prompt templates
4. Testing and refining prompts using iterative feedback loops
5. Continuously monitoring and adapting to changes in user behavior and AI model performance.

By applying the principles of Prompt Engineering, organizations can unlock the full potential of their AI systems, improve user experience, and drive business value through more effective use of conversational interfaces.

MODEL: qwen2.5:3b
Prompt Engineering is the practice of fine-tuning or customizing language models to improve their performance on specific tasks or datasets. It involves carefully selecting and crafting prompts (input text) provided to the model, in order to guide its output more accurately towards the desired result.

Here are some key aspects of Prompt Engineering:

1. Task-Specific Adaptation: Tailoring prompt engineering techniques to align with a specific task can enhance the performance of language models on that particular task.

2. Data-Driven Approaches: Often, fine-tuning is done by training a model from scratch using a modified version of the original dataset or by adding new labeled data.

3. Evaluation and Iteration: Success in prompt engineering often depends heavily on iterative refinement based on evaluation results to ensure the generated outputs are both accurate and relevant.
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Windows PowerShell

MODEL: qwen2.5:3b
Python has become very popular due to several key factors that make it a versatile and powerful language:

1. Readability: Python's syntax emphasizes readability with the use of significant whitespace. This design allows anyone who knows basic logic to read and understand code, even if they've never seen the specific programming language before.

2. Dynamic Typing: Variables do not need to be explicitly declared to specify data type and don't require a defined data type for the memory it occupies. Also, most of Python's built-in data structures (such as list, dictionary, etc.) are mutable which allows more dynamic usage in code.

3. Ease of Learning: It is considered one of the easiest programming languages to learn due to its simple syntax and clear structure.

4. Wide Application Scope: Python can be used across a range of different applications including web development (with frameworks like Django and Flask), data science, artificial intelligence, software development, automation scripting, and more.

5. Large Standard Library: The language includes an extensive standard library that supports various tasks out-of-the-box, reducing the need for developers to write their own code from scratch.

6. Extensive Ecosystem of Tools and Libraries: Python has a rich ecosystem with many libraries and tools available for specialized tasks, such as NumPy for numerical analysis, Matplotlib for plotting graphs, and TensorFlow/Keras for machine learning.

7. Open Source: Being open-source allows developers to contribute their own code back into the main library, creating a collaborative environment where bugs can be fixed and new features added continuously.

8. Strong Community Support: Python has an active and welcoming community with numerous forums, blogs, and educational resources available for learners at all levels.

9. Cross-Platform Compatibility: Python programs run on almost any platform, including Windows, macOS, Linux, Raspberry Pi, etc., which helps developers avoid compatibility issues across different environments.

10. Educational Value: Python is often used in teaching programming fundamentals due to its simplicity and the availability of online resources and tutorials.

All these factors contribute to Python's popularity as a language that can be effectively used for various purposes while offering a smooth learning curve, especially for beginners.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W5D2_Prompt_Engineering_Ollama> |
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